1. (a)

(i) Prove: if $F(x) = f(g(x))$, then $F'(x) = f'(g(x))\, g'(x)$

Since $\begin{cases} f(x+\varepsilon) = f(x) + f'(x)\varepsilon + O(\varepsilon^2) \\ g(x+\varepsilon) = g(x) + g'(x)\varepsilon + O(\varepsilon^2) \end{cases}$

$$F(x+\varepsilon) = f(g(x+\varepsilon)) = f[g(x) + g'(x)\varepsilon + O(\varepsilon^2)]$$

$$= f(g(x)) + f'(g(x))\, g'(x)\varepsilon + O(\varepsilon^2) = F(x) + F'(x)\varepsilon + O(\varepsilon^2)$$

Therefore, $F'(x) = f'(g(x))\, g'(x)$.

(ii) Prove: if $F(t) = f(x(t), y(t))$, then $F'(t) = f_x(x(t), y(t))\, x'(t) + f_y(x(t), y(t))\, y'(t)$

$$F(t+\varepsilon) = f(x(t+\varepsilon), y(t+\varepsilon)) = f(x(t) + x'(t)\varepsilon + O(\varepsilon^2),\ y(t+\varepsilon))$$

$$= f(x(t), y(t+\varepsilon)) + \partial_x f(x(t), y(t+\varepsilon))\, x'(t)\varepsilon + O(\varepsilon^2)$$

$$= f(x(t), y(t)) + \partial_y f(x(t), y(t))\, y'(t)\varepsilon + O(\varepsilon^2)$$

$$+ \partial_x f(x(t), y(t)) \cdot x'(t)\varepsilon + \partial_y \partial_x f(x(t), y(t))\, y'(t)\varepsilon \cdot x'(t)\cdot\varepsilon + O(\varepsilon^2)$$

$$= f(x(t), y(t)) + [\partial_x f(x(t), y(t))\, x'(t) + \partial_y f(x(t), y(t))\, y'(t)]\varepsilon + O(\varepsilon^2)$$

$$= F(t) + F'(t)\varepsilon + O(\varepsilon^2)$$

Therefore, $F'(t) = f_x(x(t), y(t))\, x'(t) + f_y(x(t), y(t))\, y'(t)$

1. (b)

(i) characteristic method.

Let $(x(t), y(t))$ be a line in the (x,y)-plane such that if $V(t) = u(x(t), y(t))$, then the first derivative of $u(t)$ is determined by the transport equation. Such a line $(x(t), y(t))$ is called a characteristic line. By the chain rule,

$$\frac{dV(t)}{dt} = \frac{du(x(t), y(t))}{dt} = \frac{dx(t)}{dt} u_x + \frac{dy(t)}{dt} u_y$$

Let $\frac{dx(t)}{dt} = a$, $\frac{dy(t)}{dt} = b$.

Then the line $(x(t), y(t))$ is $\begin{cases} x(t) = at + \text{constant} \\ y(t) = bt + \text{constant} \end{cases}$

which is $bx - ay = \text{constant}$.

In addition, along the line, $\frac{dV(t)}{dt} = au_x + bu_y = 0$.

Hence $u(x,y)$ is constant along each such line. The solution is $u(x,y) = f(bx - ay)$, with f an arbitrary function of one variable.

(ii) coordinate method

Let $x' = ax + by$, $y' = bx - ay$.

Replace all x and y derivatives by x' and y' derivatives.

By the chain rule, $\begin{cases} u_x = \frac{\partial x'}{\partial x} u_{x'} + \frac{\partial y'}{\partial x} \cdot u_{y'} = au_{x'} + bu_{y'} \\ u_y = \frac{\partial y'}{\partial y} u_{y'} + \frac{\partial x'}{\partial y} \cdot u_{x'} = bu_{x'} - au_{y'} \end{cases}$

Hence $au_x + bu_y = a(au_{x'} + bu_{y'}) + b(bu_{x'} - au_{y'}) = (a^2 + b^2) u_{x'}$.

Since $a^2 + b^2 \neq 0$, the equation takes the form $u_{x'} = 0$ in the new variables.

Thus, the solution is $u = f(y') = f(bx - ay)$, with f an arbitrary function of one variable.

(iii) confirm the answer by direct substitution.

The solution is $u(x,y) = f(bx-ay)$

Let $z = bx-ay$. then $u(x,y) = f(z)$

By the chain rule, $u_x = f'(z)\frac{\partial z}{\partial x} = bf'(z)$

$u_y = f'(z)\cdot\frac{\partial z}{\partial y} = -af'(z)$

$$au_x + bu_y = a(bf'(z)) + b\cdot(-af'(z)) = 0.$$

1.(c) $\begin{cases} u_x + 2xy^2 u_y = 0 \\ u(0,y) = y^2. \end{cases}$

The characteristic curves satisfy the ODE

$$\frac{dy}{dx} = \frac{2xy^2}{1} = 2xy^2.$$

$$\frac{1}{y^2}dy = 2x\,dx.$$

$$x^2 + \frac{1}{y} = C.$$

$u(x,y)$ is a constant on each such curve.

Thus $u(x,y) = f(x^2+\frac{1}{y})$ is the general solution of this PDE.

Since $u(0,y) = f(0^2+\frac{1}{y}) = f(\frac{1}{y}) = y^2$

So that $f(z) = z^{-2}$.

Therefore, $u(x,y) = f(x^2+\frac{1}{y}) = (x^2+\frac{1}{y})^{-2}$